M Sai Shanmukha Jayanth

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Education

Shiv Nadar University, Chennai, India *Aug 2023 – May 2027*
B.Tech in Computer Science and Engineering; **GPA: 8.94 (till 4th sem)**
Narayana Junior College, Hyderabad, India *July 2021 – May 2023*
Percentage: 98.6

Skills Summary

Languages:	Python, MySQL, C, Java, C++
Frameworks:	PyTorch AudioCraft,Pandas, Numpy,TensorFlow,Keras,Django,FastAPI, Scikit-Learn, Matplotlib
Specializations:	Machine Learning, Neural Networks, Deep Learning,Transformer Architecture,Audio Signal Processing,Computer Vision,Transfer Learning,Multimodal Learning,Generative AI, Backend Development,Image Classification .
Platforms:	Google Colab, Jupyter Notebook, VS Code, Git, GitHub
Soft Skills:	Communication, Teamwork, Leadership, Problem Solving

Work Experience

Machine Learning Intern — Labmentix *July 2025 – Jan 2026*

- Built end-to-end machine learning pipelines across domains such as retail analytics, logistics, real estate, and customer feedback, covering **data preprocessing, feature engineering, model training, and evaluation** using Python and Scikit-learn.
- Developed predictive models **(Random Forest, Gradient Boosting, Linear Models, XGBoost)** for tasks like delivery time prediction, customer satisfaction analysis, and investment insights, applying cross-validation and hyperparameter tuning.
- Engineered complex features including **geospatial distance calculations, time-based variables, sentiment extraction from text, and multi-dataset joins**, improving model performance on real-world datasets. deep learning computer vision models **(CNNs with transfer learning)** for aerial object classification, leveraging pretrained architectures for improved accuracy.

Projects

Aerial Object Detection *Sept 2025 – Sept 2025*
Project Link

- Built an end-to-end **deep learning pipeline** for aerial image classification, covering preprocessing, augmentation, training, and evaluation.
- Improved model performance from **$\tilde{5}$6% to $\tilde{9}$7% accuracy** using **MobileNetV2 transfer learning**, demonstrating effective feature reuse over a baseline CNN.
- Implemented **efficient data pipelines (tf.data, caching, prefetching)** and augmentation strategies to enhance generalization and training stability.
- Evaluated models using **confusion matrix, precision, recall, and F1-score**, and performed **comparative analysis** to select the optimal architecture.

Restaurant Clustering Analysis *July 2025 – Aug 2025*
Project Link

- **Engineered** and cleaned a unified dataset; used **VADER** and **TF-IDF** for **NLP** sentiment scoring.
- Validated insights via **t-tests** and **Correlation Analysis**, identifying sentiment and service as key drivers.
- Developed and optimized K-Means model on features reduced by PCA, retaining 95% variance.
- Delivered **actionable segments** (validated via **Silhouette Score**) for marketing strategy.

Integrated Retail Analysis *Aug 2025 – Sept 2025*
Project Link

- **Engineered high-impact time-series features** (Lagged Scales and Rolling Mean) to capture sales memory and short-term trends for model accuracy.
- **Developed an XGBoost Regressor** leveraging economic and promotional data to forecast weekly sales at the store-department level.
- **Delivered a robust predictive model** that directly supports retail **inventory optimization** and **staffing allocation** for cost efficiency.

Certificates

- **Supervised Machine Learning: Regression and Classification (Coursera)** – May 2025: building and evaluating predictive models.
- **Neural Networks and Deep Learning (Coursera)** – June 2025: practical experience in neural network design and training.
- **Improving Deep Neural Networks: Hyperparameter Tuning, Regularization, and Optimization (Coursera)** – July 2025: optimizing neural network performance.